AtliQo Bank Credit Card Launch: Phase 1

Objective: Analyze customers' transactions and credit profiles to figure out a target group for the launch of AtliQo bank credit card

In [309…
```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
import warnings
warnings.filterwarnings("ignore")
```

Data Import

We got the dataset in two formats (1) CSV (2) MySQL. I will show you how to import data from both. You can use only one method out of these two.

Option 1. Read data from CSV

In [310…
```
df_cust = pd.read_csv('data/customers.csv')
df_cs = pd.read_csv('data/credit_profiles.csv')
df_trans = pd.read_csv('data/transactions.csv')
```

Option 2. Read it from MySQL

In [311…
```
import mysql.connector

conn = mysql.connector.connect(
    host='localhost',
    user='root',
    passwd='root',
    database='e_master_card'
)

df_cust = pd.read_sql("SELECT * FROM customers", conn)
df_cust.head(3)
```

Out[311…

	cust_id	name	gender	age	location	occupation	annual_income	marital_status
0	1	Dhruv Bansal	Female	2	City	Business Owner	358211.0	Married
1	2	Kabir Thakur	Female	47	City	Consultant	65172.0	Single
2	3	Kriti Mangal	Male	21	City	Freelancer	22378.0	Married

In [312…
```
df_trans = pd.read_sql("SELECT * FROM transactions", conn)
df_trans.head(3)
```

Out[312…

	tran_id	cust_id	tran_date	tran_amount	platform	product_category	payment_type
0	1	705	2023-01-01	63	Flipkart	Electronics	Phonepe
1	2	385	2023-01-01	99	Alibaba	Fashion & Apparel	Credit Card
2	3	924	2023-01-01	471	Shopify	Sports	Phonepe

In [313…
```
df_cs = pd.read_sql("SELECT * FROM credit_profiles", conn)
df_cs.head(3)
```

Out[313…

	cust_id	credit_score	credit_utilisation	outstanding_debt	credit_inquiries_last_6_months	credit_limit
0	1	749	0.585171	19571.0	0.0	40000.0
1	2	587	0.107928	161644.0	2.0	1250.0
2	3	544	0.854807	513.0	4.0	1000.0

In [314…
```
# when you are done importing the data, close the connection
conn.close()
```

In [315…
```
print("Customers data",df_cust.shape)
print("Credit Score data",df_cs.shape)
print("Transactions data",df_trans.shape)
```

```
Customers data (1000, 8)
Credit Score data (1004, 6)
Transactions data (500000, 7)
```

In [316…
```
df_cust.head()
```

Out[316…

	cust_id	name	gender	age	location	occupation	annual_income	marital_status
0	1	Dhruv Bansal	Female	2	City	Business Owner	358211.0	Married
1	2	Kabir Thakur	Female	47	City	Consultant	65172.0	Single
2	3	Kriti Mangal	Male	21	City	Freelancer	22378.0	Married
3	4	Sara Dwivedi	Male	24	Rural	Freelancer	33563.0	Married
4	5	Aarohi Negi	Male	48	City	Consultant	39406.0	Married

In [317…

```
df_cs.head()
```

Out[317…

	cust_id	credit_score	credit_utilisation	outstanding_debt	credit_inquiries_last_6_months	credit_limit
0	1	749	0.585171	19571.0	0.0	40000.0
1	2	587	0.107928	161644.0	2.0	1250.0
2	3	544	0.854807	513.0	4.0	1000.0
3	4	504	0.336938	224.0	2.0	1000.0
4	5	708	0.586151	18090.0	2.0	40000.0

In [318…

```
df_trans.head()
```

Out[318…

	tran_id	cust_id	tran_date	tran_amount	platform	product_category	payment_type
0	1	705	2023-01-01	63	Flipkart	Electronics	Phonepe
1	2	385	2023-01-01	99	Alibaba	Fashion & Apparel	Credit Card
2	3	924	2023-01-01	471	Shopify	Sports	Phonepe
3	4	797	2023-01-01	33	Shopify	Fashion & Apparel	Gpay
4	5	482	2023-01-01	68	Amazon	Fashion & Apparel	Net Banking

Explore Customers Table

In [319…

```
df_cust.head(3)
```

Out[319…

	cust_id	name	gender	age	location	occupation	annual_income	marital_status
0	1	Dhruv Bansal	Female	2	City	Business Owner	358211.0	Married
1	2	Kabir Thakur	Female	47	City	Consultant	65172.0	Single
2	3	Kriti Mangal	Male	21	City	Freelancer	22378.0	Married

In [320…

```
df_cust.describe()
```

Out[320…

	cust_id	age	annual_income
count	1000.000000	1000.000000	950.000000
mean	500.500000	36.405000	139410.314737
std	288.819436	15.666155	112416.802007
min	1.000000	1.000000	2.000000
25%	250.750000	26.000000	47627.500000
50%	500.500000	32.000000	112218.500000
75%	750.250000	46.000000	193137.500000
max	1000.000000	135.000000	449346.000000

1. Analyze Income Column

Handle Null Values: Annual income

Now let us check if any of our dataframe columns contain null values

In [321…

```
df_cust.isnull().sum()
```

Out[321…
```
cust_id            0
name               0
gender             0
age                0
location           0
occupation         0
annual_income     50
marital_status     0
dtype: int64
```

Ahh.. 50 null values in annual_income. Let's quickly explore those rows

In [322…
```
df_cust[df_cust.annual_income.isna()].head(4)
```

Out[322…

	cust_id	name	gender	age	location	occupation	annual_income	marital_status
14	15	Isha Arora	Female	25	Rural	Artist	NaN	Married
82	83	Sanjana Singhal	Male	27	City	Freelancer	NaN	Single
97	98	Sia Batra	Male	47	Suburb	Business Owner	NaN	Married
102	103	Pooja Ganguly	Male	32	City	Data Scientist	NaN	Married

We can handle these null values using different ways,

1. **Remove them**: Since there are 50 of them in a dataframe of 1000, we will not remove them as we don't want to loose some important records
2. **Replace them with mean or median**: It is suggested with use median in the case of income. This is because in an income data there could be outliers and median is more robust to these outliers
3. **Replace them with median per occupation**: Occupation wise median income can vary. It is best to use a median per occupation for replacement

In [323…
```
occupation_wise_inc_median = df_cust.groupby("occupation")["annual_income"].median()
occupation_wise_inc_median
```

Out[323…
```
occupation
Accountant              65265.0
Artist                  45794.0
Business Owner         261191.5
Consultant              58017.0
Data Scientist         135759.0
Freelancer              46759.0
Fullstack Developer     76774.0
Name: annual_income, dtype: float64
```

In [324…
```
occupation_wise_inc_median['Artist']
```

Out[324…
```
45794.0
```

In [325…
```
# 2. Replace null values in annual_income with the median income of their occupation group
df_cust['annual_income'] = df_cust.apply(
    lambda row: occupation_wise_inc_median[row['occupation']] if pd.isnull(row['annual_income']) else row['annual_income'],
    axis=1
)
```

In [326…
```
df_cust.iloc[[1,29]]
```

Out[326…

	cust_id	name	gender	age	location	occupation	annual_income	marital_status
1	2	Kabir Thakur	Female	47	City	Consultant	65172.0	Single
29	30	Suhani Choudhary	Male	31	Rural	Data Scientist	105583.0	Married

Previously records at location 1 and 29 had null annual income. Now you have a median value per occupation

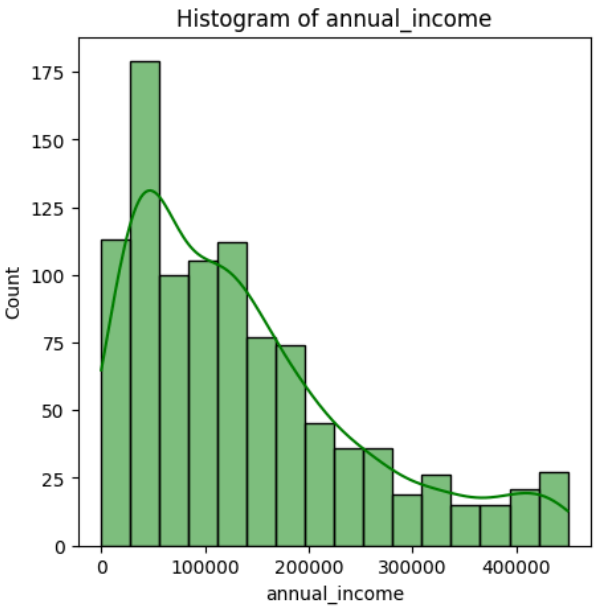
In [327…
```
df_cust.isnull().sum()
```

Out[327…
```
cust_id            0
name               0
gender             0
age                0
location           0
occupation         0
annual_income      0
marital_status     0
dtype: int64
```

Number of null values in all the columns is zero now!

Now that there are no null values, let us view the distribution of annual income

In [328…
```
plt.figure(figsize=(5, 5))
sns.histplot(df_cust['annual_income'], kde=True, color='green', label='Data')
plt.title('Histogram of annual_income')
plt.show()
```



You can see above that the income distribution is right skewed

Let us now use describe() function to check some quick stats

In [329…
```
df_cust.describe()
```

Out[329…

	cust_id	age	annual_income
count	1000.000000	1000.000000	1000.000000
mean	500.500000	36.405000	138916.765500
std	288.819436	15.666155	110969.408643
min	1.000000	1.000000	2.000000
25%	250.750000	26.000000	48229.500000
50%	500.500000	32.000000	113416.000000
75%	750.250000	46.000000	192614.000000
max	1000.000000	135.000000	449346.000000

We have following observations from the above,

1. **Age**: min = 1, max = 135
2. **Annual Income**: min = 2, max = 447 k

Age column has outliers. Annual income also seem to have outliers in terms of minimum value because business suggested that minimum income should be atleast 100

In [330…
```
df_cust.annual_income.describe()
```

Out[330…
```
count      1000.000000
mean     138916.765500
std      110969.408643
min           2.000000
25%       48229.500000
50%      113416.000000
75%      192614.000000
max      449346.000000
Name: annual_income, dtype: float64
```

Outlier Detection: Annual income

Let us use standard deviation to detect outliers. Common practice is to treat anything that +/- 3 std dev as an outlier

In [331…
```
df_cust['annual_income'].mean(), df_cust['annual_income'].std()
```

Out[331…
```
(138916.7655, 110969.4086433663)
```

In [332…
```
lower = df_cust['annual_income'].mean() - 3*df_cust['annual_income'].std()
upper = df_cust['annual_income'].mean() + 3*df_cust['annual_income'].std()

lower, upper
```

Out[332…
```
(-193991.4604300989, 471824.9914300989)
```

In [333…
```
df_cust[df_cust['annual_income']>upper]
```

Out[333…

cust_id	name	gender	age	location	occupation	annual_income	marital_status

We are seeing two outliers as per our statistical criteria of +/3 3 std deb. But we don't always assume these as outliers all the time. We have to use business knowledge and our sense of judgement. Here after discussing with the business we concluded that having this type of higher income for business owners is usual and we will keep these data points as is to stay close to the reality while doing our analysis.

On the lower end however, we see minimum income as 2. Our business manager has told us that the income should be at least 100. We can use this as our criteria to find out the outliers on the lower end. These outliers could have occured due to a data error.

In [334…
```
df_cust[df_cust.annual_income<100]
```

Out[334…

	cust_id	name	gender	age	location	occupation	annual_income	marital_status
31	32	Veer Gambhir	Male	50	City	Business Owner	50.0	Married
262	263	Veer Rathod	Male	53	Suburb	Business Owner	50.0	Married
316	317	Avinash Chauhan	Male	47	City	Consultant	50.0	Married
333	334	Samaira D'Souza	Female	29	City	Data Scientist	50.0	Married
340	341	Ridhi Rastogi	Female	33	Rural	Fullstack Developer	50.0	Married
543	544	Gauri Gupta	Male	54	City	Consultant	2.0	Married
592	593	Vivaan Pande	Female	32	City	Business Owner	50.0	Married
633	634	Aradhya Pandey	Male	26	City	Data Scientist	2.0	Married
686	687	Gauri Nanda	Male	40	City	Business Owner	2.0	Married
696	697	Avani Lalwani	Male	47	City	Consultant	20.0	Married

Outlier Treatment: Annual income

Above records (with <100$ income) are outliers. We have following options to treat them,

1. **Remove them**: After discussion with business, we decided not to remove them as these are valid customers and we want to include them in our analysis
2. **Replace them with mean or median** : Mean is sensitive to outliers. It is better to use median for income values
3. **Replace them with occupation wise median**: Income level may vary based on occupation. For example median income for data scientist can be different from a median income of a business owner. It is better to use occupation wise median income for replacement

In [335…
```
occupation_wise_inc_median["Artist"]
```

Out[335…
```
45794.0
```

In [336…
```
for index, row in df_cust.iterrows():
    if row["annual_income"] < 100:
        occupation = df_cust.at[index, "occupation"]
        df_cust.at[index, "annual_income"] = occupation_wise_inc_median[occupation]
```

In [337…
```
df_cust[df_cust.annual_income<100]
```

Out[337…

cust_id	name	gender	age	location	occupation	annual_income	marital_status

In [338…
```
df_cust.loc[[112,256]]
```

Out[338…

	cust_id	name	gender	age	location	occupation	annual_income	marital_status
112	113	Ishita Mehrotra	Male	55	City	Business Owner	303207.0	Married
256	257	Avinash Jain	Male	28	City	Freelancer	205791.0	Married

Record at 112 and 256 location had annual income of < 100$. Now you can see it is replaced by a median income per occupation

Data Visualization: Annual Income

We will explore average income level based on occupation, gender, location and marital status

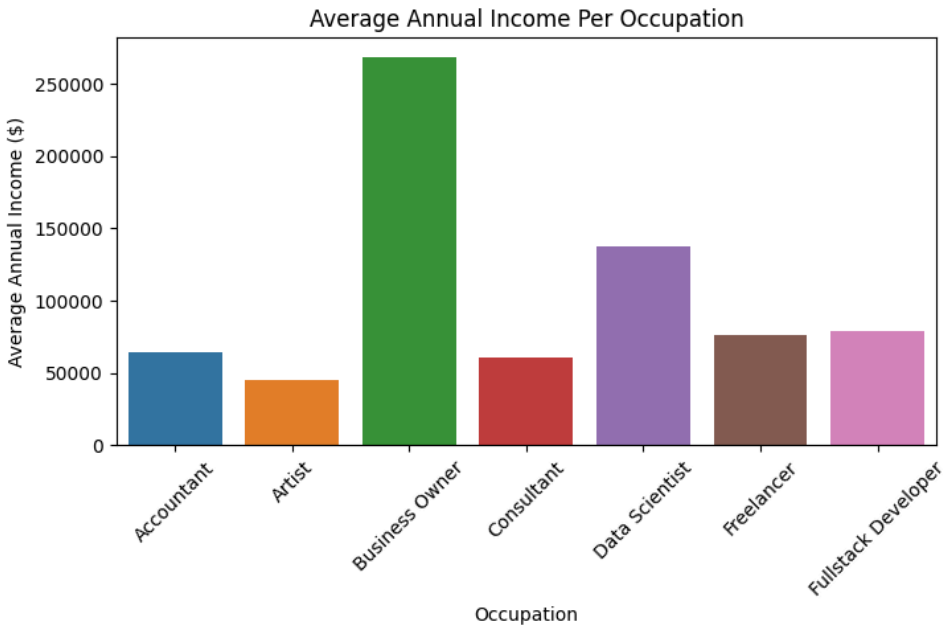
In [339…
```
avg_income_per_occupation  = df_cust.groupby("occupation")["annual_income"].mean()
avg_income_per_occupation
```

Out[339…

```
occupation
Accountant              64123.562500
Artist                  45309.236842
Business Owner         268447.368512
Consultant              60703.154639
Data Scientist         137021.266304
Freelancer              76327.508772
Fullstack Developer     78727.972973
Name: annual_income, dtype: float64
```

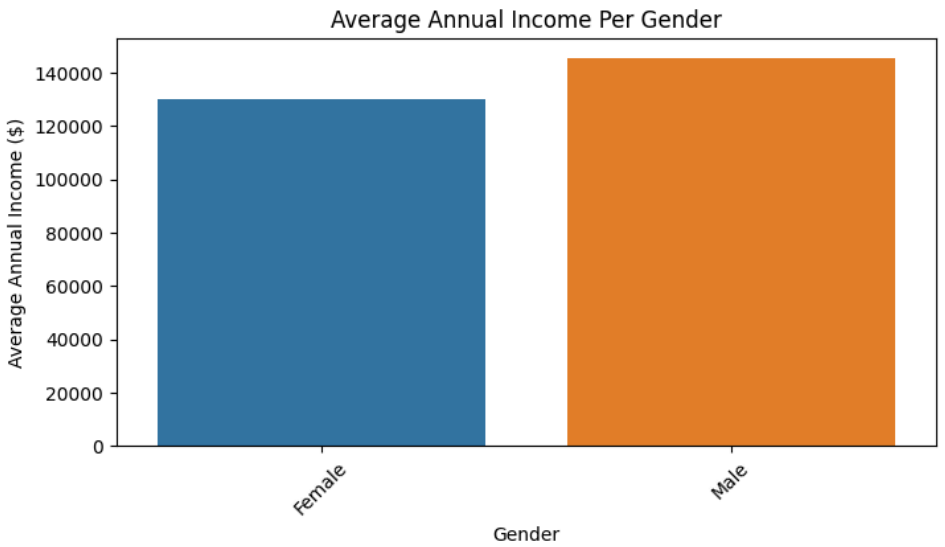
In [340…

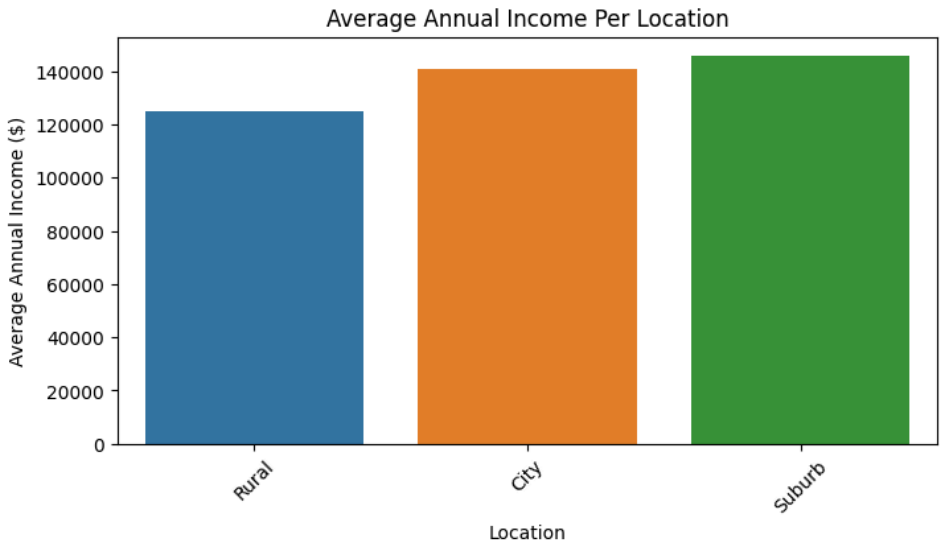
```
plt.figure(figsize=(8,4))
sns.barplot(x=avg_income_per_occupation.index, y=avg_income_per_occupation.values, palette='tab10')
plt.xticks(rotation=45)
plt.title('Average Annual Income Per Occupation')
plt.xlabel('Occupation')
plt.ylabel('Average Annual Income ($)')
plt.show()
```

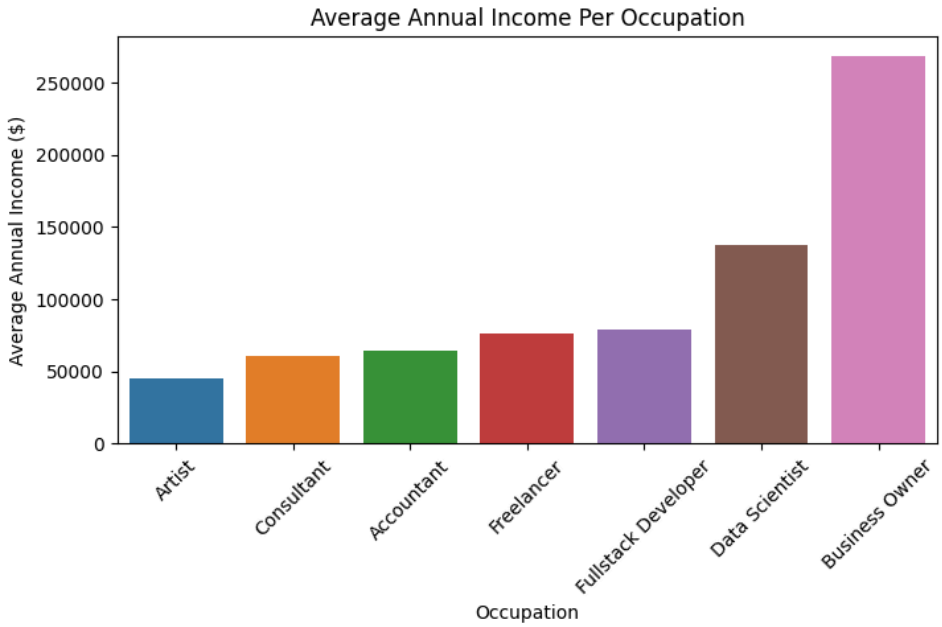


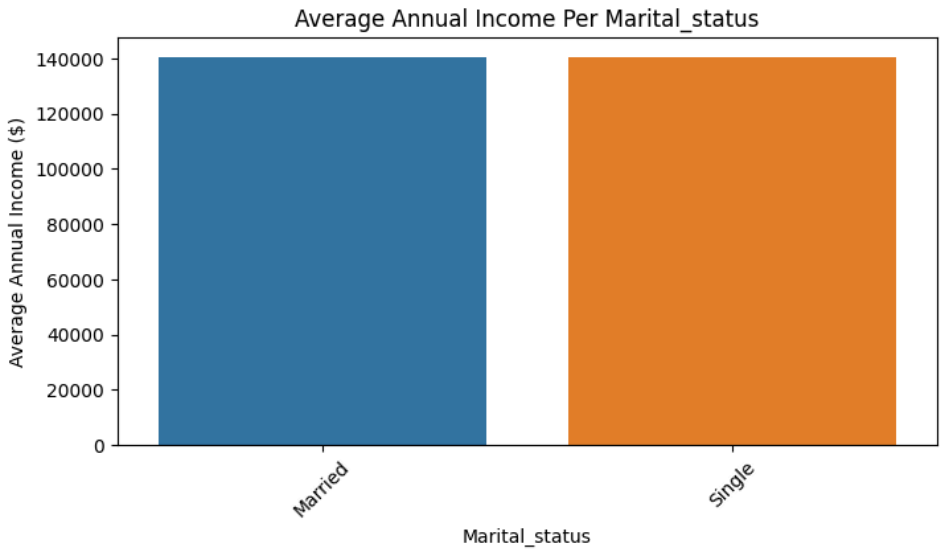
In [341…

```
# List of categorical columns
categorical_columns = ['gender', 'location', 'occupation', 'marital_status']

# Loop through each categorical column and plot a bar chart of average annual income
for col in categorical_columns:
    plt.figure(figsize=(8, 4))
    avg_income_per_group = df_cust.groupby(col)['annual_income'].mean().sort_values()
    sns.barplot(x=avg_income_per_group.index, y=avg_income_per_group.values, palette='tab10')
    plt.xticks(rotation=45)
    plt.title(f'Average Annual Income Per {col.capitalize()}')
    plt.xlabel(col.capitalize())
    plt.ylabel('Average Annual Income ($)')
    plt.show()
```





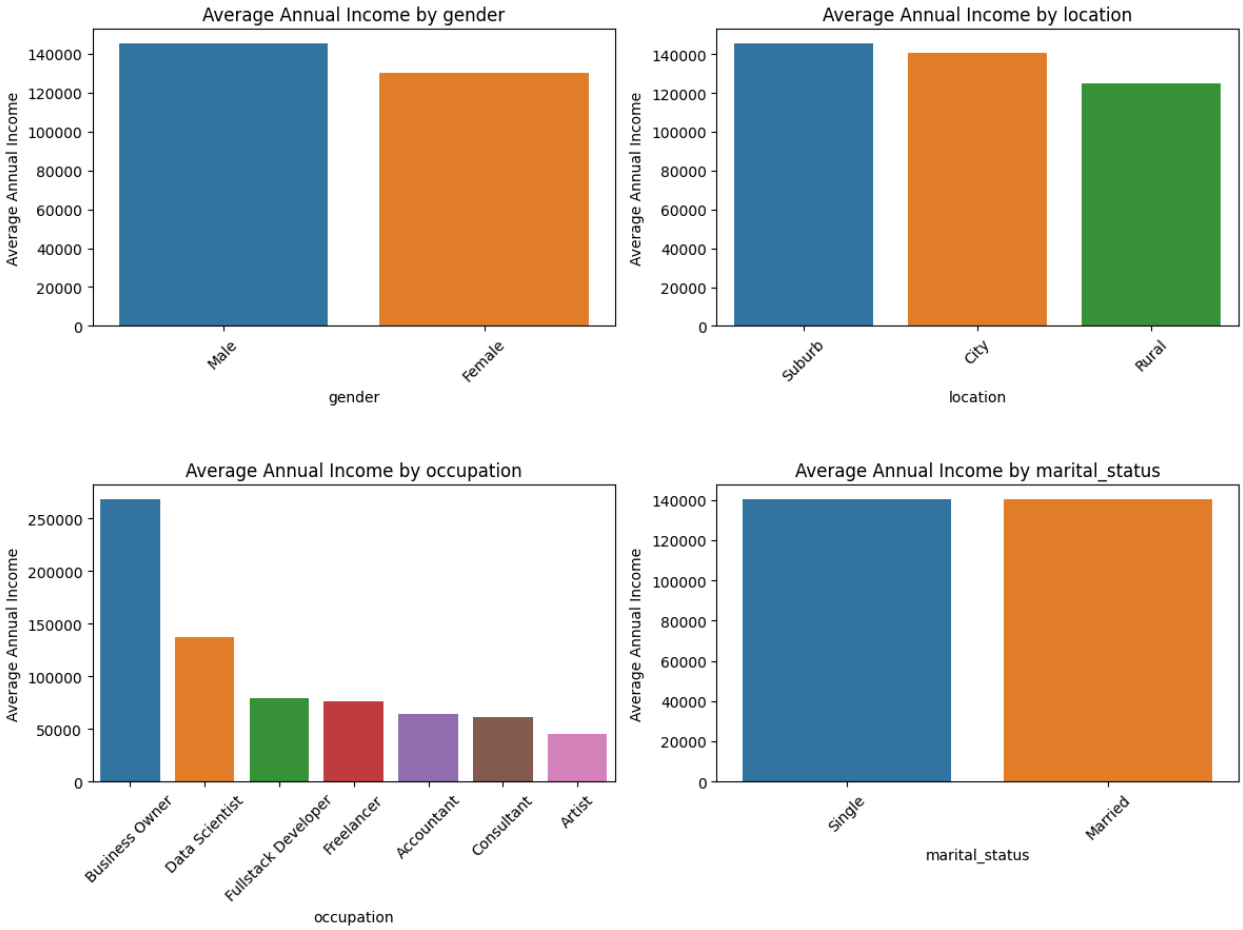




In [342…

```
# List of categorical columns
cat_cols = ['gender', 'location', 'occupation', 'marital_status']

num_rows = 3
# Create subplots
fig, axes = plt.subplots(num_rows, 2, figsize=(12, 4 * num_rows))

# Flatten the axes array to make it easier to iterate
axes = axes.flatten()

# Create subplots for each categorical column
```

```
for i, cat_col in enumerate(cat_cols):
    # Calculate the average annual income for each category
    avg_income_by_category = df_cust.groupby(cat_col)['annual_income'].mean().reset_index()

    # Sort the data by 'annual_income' before plotting
    sorted_data = avg_income_by_category.sort_values(by='annual_income', ascending=False)

    sns.barplot(x=cat_col, y='annual_income', data=sorted_data, ci=None, ax=axes[i], palette='tab10')
    axes[i].set_title(f'Average Annual Income by {cat_col}')
    axes[i].set_xlabel(cat_col)
    axes[i].set_ylabel('Average Annual Income')

    # Rotate x-axis labels for better readability
    axes[i].set_xticklabels(axes[i].get_xticklabels(), rotation=45)

# Hide any unused subplots
for i in range(len(cat_cols), len(axes)):
    fig.delaxes(axes[i])
plt.tight_layout()
plt.show()
```



2. Analyze Age Column

Handle Null Values: Age Column

First let us check if there are any NULL values in the Age column

In [343…

```
df_cust.age.isnull().sum()
```

Out[343…

```
0
```

No null values are found in age column. This means we don't need to worry about handling them.

In [344…

```
df_cust.describe()
```

Out[344…

	cust_id	age	annual_income
count	1000.000000	1000.000000	1000.000000
mean	500.500000	36.405000	140483.548500
std	288.819436	15.666155	110463.002934
min	1.000000	1.000000	5175.000000
25%	250.750000	26.000000	49620.500000
50%	500.500000	32.000000	115328.000000
75%	750.250000	46.000000	195514.250000
max	1000.000000	135.000000	449346.000000

Outlier Treatment: Age

Above we see that min age is 1 and max age is 135. These seem to be outliers. So let's find out age distribution.

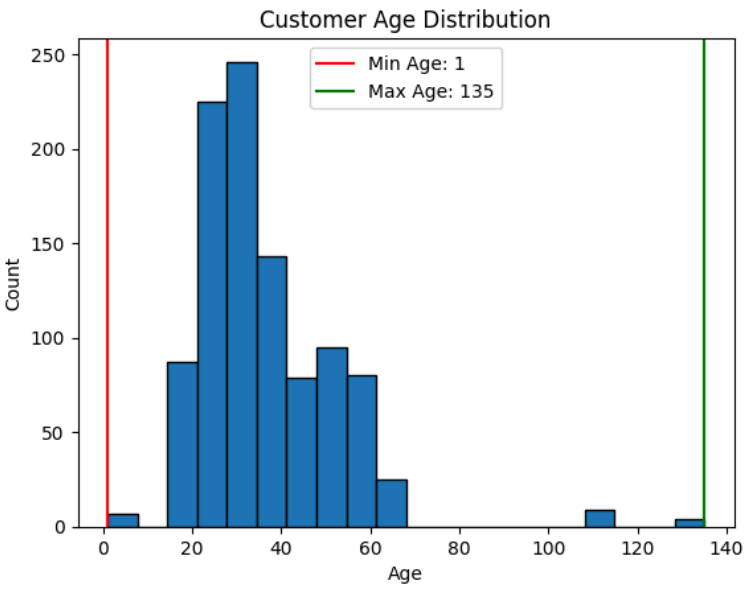
In [345…

```
min_age = df_cust.age.min()
max_age = df_cust.age.max()

min_age, max_age
```

Out[345… (1, 135)

In [346…

```
plt.hist(df_cust.age, bins=20, edgecolor='black')
plt.xlabel("Age")
plt.ylabel("Count")
plt.title("Customer Age Distribution")

plt.axvline(min_age, color="red", label=f"Min Age: {min_age}")
plt.axvline(max_age, color="green", label=f"Max Age: {max_age}")

plt.legend()
plt.show()
```



From above we will try to find out all customers above 80 and below 15.

In [347…

```
df_cust[(df_cust.age<15)|(df_cust.age>80)]
```

Out[347…

	cust_id	name	gender	age	location	occupation	annual_income	marital_status
0	1	Dhruv Bansal	Female	2	City	Business Owner	358211.0	Married
41	42	Manya Thakur	Male	110	City	Artist	7621.0	Married
165	166	Swara Mehtani	Female	1	City	Freelancer	39721.0	Single
174	175	Sneha Jaiswal	Male	110	City	Freelancer	23723.0	Married
222	223	Aaryan Bajaj	Male	110	Suburb	Freelancer	210987.0	Married
277	278	Amaira Dewan	Male	110	City	Consultant	96522.0	Single
295	296	Veer Anand	Male	1	Rural	Accountant	55254.0	Married
325	326	Yash Chatterjee	Male	110	City	Accountant	61021.0	Single
610	611	Advait Mangal	Male	135	Rural	Business Owner	444776.0	Married
692	693	Aarohi Dubey	Male	1	City	Business Owner	83045.0	Married
703	704	Aradhya Goswami	Female	110	City	Freelancer	43404.0	Single
709	710	Rehan Patel	Female	110	City	Data Scientist	98417.0	Married
728	729	Yuvraj Gambhir	Male	135	City	Business Owner	382836.0	Married
832	833	Pranav Dwivedi	Female	110	City	Fullstack Developer	95379.0	Single
845	846	Anika Kadam	Male	1	City	Consultant	20838.0	Married
855	856	Aarav D'Souza	Female	2	City	Fullstack Developer	30689.0	Married
895	896	Ritvik Soni	Male	1	City	Freelancer	31533.0	Married
923	924	Kriti Joshi	Male	110	City	Freelancer	51629.0	Married
951	952	Ishan Khanna	Male	135	City	Data Scientist	49677.0	Married
991	992	Krishna Banerjee	Male	135	City	Fullstack Developer	93267.0	Single

In [348…

```
outliers = df_cust[(df_cust.age<15)|(df_cust.age>80)]
outliers.shape
```

Out[348…

```
(20, 8)
```

Total 20 outliers for age. Now how can we handle these outliers?

Possible options,

1. Remove them: This doesn't sound like a good option as we will loose important information
2. Replace outlier values with some appropriate value: We can use mean or median for this

In [349…

```
df_cust.age.median()
```

Out[349…

```
32.0
```

Instead of replace it with a median age for all customers, how about we calculate median age per occupation?

In [350…

```
outliers.head(3)
```

Out[350…

	cust_id	name	gender	age	location	occupation	annual_income	marital_status
0	1	Dhruv Bansal	Female	2	City	Business Owner	358211.0	Married
41	42	Manya Thakur	Male	110	City	Artist	7621.0	Married
165	166	Swara Mehtani	Female	1	City	Freelancer	39721.0	Single

As you can see, for business owners median age is 49 whereas artists have youngest age

We will calculte median per occupation and then use that for replacing outliers

In [351…

```
median_age_per_occupation = df_cust.groupby('occupation')['age'].median()
median_age_per_occupation
```

Out[351…

```
occupation
Accountant             31.5
Artist                 26.0
Business Owner         51.0
Consultant             46.0
Data Scientist         32.0
Freelancer             24.0
Fullstack Developer    27.5
Name: age, dtype: float64
```

In [352…

```
for index, row in outliers.iterrows():
    if pd.notnull(row['age']):
        occupation = df_cust.at[index, 'occupation']
        df_cust.at[index, 'age'] = median_age_per_occupation[occupation]
```

In [353…
```
df_cust[(df_cust.age<15)|(df_cust.age>80)]
```

Out[353…

cust_id	name	gender	age	location	occupation	annual_income	marital_status

In [354…
```
df_cust.age.describe()
```

Out[354…
```
count    1000.000000
mean       35.541500
std        12.276634
min        18.000000
25%        26.000000
50%        32.000000
75%        44.250000
max        64.000000
Name: age, dtype: float64
```

As you can see above, now we don't have any outliers left. min age is 18 and max is 64

In [355…
```
df_cust.head()
```

Out[355…

	cust_id	name	gender	age	location	occupation	annual_income	marital_status
0	1	Dhruv Bansal	Female	51.0	City	Business Owner	358211.0	Married
1	2	Kabir Thakur	Female	47.0	City	Consultant	65172.0	Single
2	3	Kriti Mangal	Male	21.0	City	Freelancer	22378.0	Married
3	4	Sara Dwivedi	Male	24.0	Rural	Freelancer	33563.0	Married
4	5	Aarohi Negi	Male	48.0	City	Consultant	39406.0	Married

Data Visualization: Age Column

In [356…
```
# Define the bin edges and labels
bin_edges = [17, 25, 48, 65]  # Adjust as needed
bin_labels = ['18-25', '26-48', '49-65']

# Use the cut function to bin and label the age column
pd.cut(df_cust['age'], bins=bin_edges, labels=bin_labels)
```

Out[356…
```
0      49-65
1      26-48
2      18-25
3      18-25
4      26-48
       ...
995    26-48
996    49-65
997    26-48
998    26-48
999    26-48
Name: age, Length: 1000, dtype: category
Categories (3, object): ['18-25' < '26-48' < '49-65']
```

In [357…
```
# Define the bin edges and labels
bin_edges = [17, 25, 48, 65]  # Adjust as needed
bin_labels = ['18-25', '26-48', '49-65']

# Use the cut function to bin and label the age column
df_cust['age_group'] = pd.cut(df_cust['age'], bins=bin_edges, labels=bin_labels)
```

In [358…
```
df_cust.head()
```

Out[358…

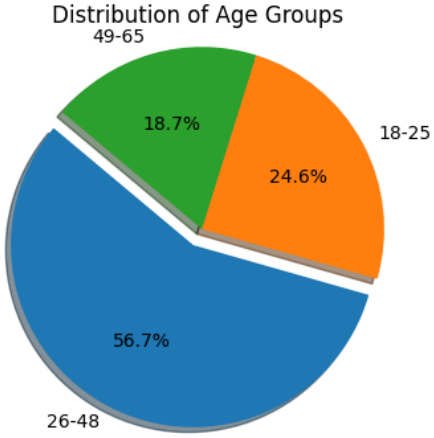
	cust_id	name	gender	age	location	occupation	annual_income	marital_status	age_group
0	1	Dhruv Bansal	Female	51.0	City	Business Owner	358211.0	Married	49-65
1	2	Kabir Thakur	Female	47.0	City	Consultant	65172.0	Single	26-48
2	3	Kriti Mangal	Male	21.0	City	Freelancer	22378.0	Married	18-25
3	4	Sara Dwivedi	Male	24.0	Rural	Freelancer	33563.0	Married	18-25
4	5	Aarohi Negi	Male	48.0	City	Consultant	39406.0	Married	26-48

In [359…
```
df_cust['age_group'].value_counts(normalize=True)*100
```

Out[359…
```
age_group
26-48    56.7
18-25    24.6
49-65    18.7
Name: proportion, dtype: float64
```

In [360…
```
# Calculate the count of values in each age group
age_group_counts = df_cust['age_group'].value_counts(normalize=True) * 100

# Plot the pie chart
```

```
plt.figure(figsize=(4, 4))
plt.pie(
    age_group_counts,
    labels=age_group_counts.index,
    explode=(0.1,0,0),
    autopct='%1.1f%%',
    shadow=True,
    startangle=140)
plt.axis('equal')  # Equal aspect ratio ensures that pie is drawn as a circle.
plt.title('Distribution of Age Groups')
plt.show()
```

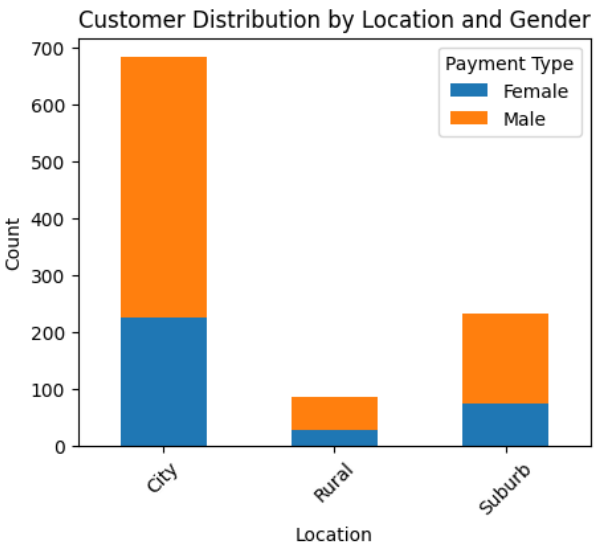


More than 50% of customer base are in in age group of 26 - 48 adn ~26% are of age group 18 - 25

3. Analyze Gender and Location Distribution

In [361…

```
customer_location_gender = df_cust.groupby(['location', 'gender']).size().unstack(fill_value=0)

# Create a stacked bar chart to visualize the distribution of payment types for each occupation
customer_location_gender.plot(kind='bar', stacked=True, figsize=(5, 4))

# Add labels and title
plt.xlabel('Location')
plt.ylabel('Count')
plt.title('Customer Distribution by Location and Gender')

# Show the bar chart
plt.legend(title='Payment Type', bbox_to_anchor=(1, 1))  # Add a legend

# Rotate the x-axis labels for better readability
plt.xticks(rotation=45)

plt.show()
```



Explore Credit Score Table

In [362…

```
df_cs.head()
```

Out[362…

	cust_id	credit_score	credit_utilisation	outstanding_debt	credit_inquiries_last_6_months	credit_limit
0	1	749	0.585171	19571.0	0.0	40000.0
1	2	587	0.107928	161644.0	2.0	1250.0
2	3	544	0.854807	513.0	4.0	1000.0
3	4	504	0.336938	224.0	2.0	1000.0
4	5	708	0.586151	18090.0	2.0	40000.0

Data Cleaning Step 1: Remove Duplicates

In [363…

```
df_cs.shape
```

Out[363…

```
(1004, 6)
```

Hmmm... there are 1004 rows in this dataframe whereas customers dataframe had only 1000. There might be invalid or duplicate data in df_cs

In [364…

```
df_cs['cust_id'].nunique()
```

Out[364…

```
1000
```

In [365…

```
df_cs.duplicated('cust_id')
```

Out[365…

```
0       False
1       False
2       False
3       False
4       False
        ...
999     False
1000    False
1001    False
1002    False
1003    False
Length: 1004, dtype: bool
```

In [366…

```
df_cs[df_cs.duplicated('cust_id', keep=False)]
```

Out[366…

	cust_id	credit_score	credit_utilisation	outstanding_debt	credit_inquiries_last_6_months	credit_limit
516	517	308	NaN	NaN	NaN	NaN
517	517	308	0.113860	33.0	3.0	500.0
569	569	344	NaN	NaN	NaN	NaN
570	569	344	0.112599	37.0	0.0	500.0
607	606	734	NaN	NaN	NaN	NaN
608	606	734	0.193418	4392.0	1.0	40000.0
664	662	442	NaN	NaN	NaN	NaN
665	662	442	0.856039	266.0	2.0	500.0

In [367…

```
df_cs_clean_1 = df_cs.drop_duplicates(subset='cust_id', keep="last")
df_cs_clean_1.shape
```

Out[367…

```
(1000, 6)
```

In [368…

```
df_cs_clean_1[df_cs_clean_1.duplicated('cust_id', keep=False)]
```

Out[368…

cust_id	credit_score	credit_utilisation	outstanding_debt	credit_inquiries_last_6_months	credit_limit

df_cs_clean_1 looks clean now after cleaning duplicates.

Next step would be to see if there are any null values

Data Cleaning Step 2: Handle Null Values

In [369…

```
df_cs_clean_1.isnull().sum()
```

Out[369…

```
cust_id                           0
credit_score                      0
credit_utilisation                0
outstanding_debt                  0
credit_inquiries_last_6_months    0
credit_limit                     65
dtype: int64
```

Ahh... look at credit_limit. It has a bunch of null values. we need to clean them up! From the business knowledge we know that credit limit depends on credit score of a customer. We will try to find out if we can figure out a mathematical relationship between credit score and credit limit and use

credit score to full NULL values in credit limit. Let's explore a few things here!

In [370…
```
df_cs_clean_1[df_cs_clean_1.credit_limit.isnull()]
```

Out[370…

	cust_id	credit_score	credit_utilisation	outstanding_debt	credit_inquiries_last_6_months	credit_limit
10	11	679	0.557450	9187.0	2.0	NaN
35	36	790	0.112535	4261.0	1.0	NaN
37	38	514	0.296971	238.0	2.0	NaN
45	46	761	0.596041	24234.0	2.0	NaN
64	65	734	0.473715	13631.0	0.0	NaN
...	...	...	...	...	...	...
912	909	479	0.487555	320.0	3.0	NaN
931	928	311	0.832244	316.0	2.0	NaN
948	945	526	0.272734	227.0	1.0	NaN
954	951	513	0.175914	131.0	3.0	NaN
957	954	783	0.867421	46451.0	0.0	NaN

65 rows × 6 columns

In [371…
```
df_cs_clean_1['credit_limit'].unique()
```

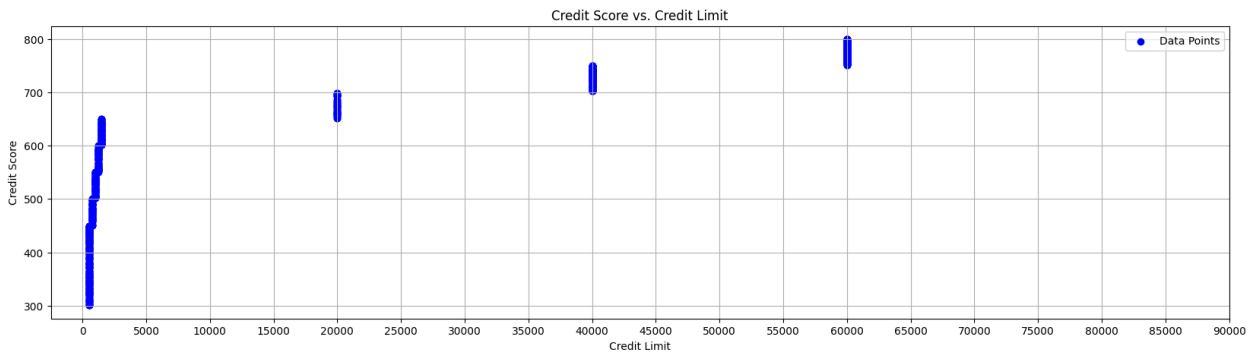
Out[371…
```
array([40000.,  1250.,  1000.,   500.,   750.,     nan,  1500., 60000.,
       20000.])
```

Credit limit has only few unique values. Let's check the count for each of these unique values

In [372…
```
df_cs_clean_1['credit_limit'].value_counts()
```

Out[372…
```
credit_limit
500.0      229
60000.0    186
40000.0    137
1500.0     100
1000.0      90
750.0       76
1250.0      75
20000.0     42
Name: count, dtype: int64
```

In [373…
```
# Looking at scatter plot for credit score vs credit_limit again (after handling oultiers)
# Create a scatter plot
plt.figure(figsize=(20, 5))
plt.scatter(df_cs_clean_1['credit_limit'], df_cs_clean_1['credit_score'], c='blue', marker='o', label='Data Points')

# Customize the plot
plt.title('Credit Score vs. Credit Limit')
plt.xlabel('Credit Limit')
plt.ylabel('Credit Score')

# Adjust the y-axis bin interval to 1000
plt.xticks(range(0, 90001, 5000))
plt.grid(True)

# Show the plot
plt.legend()
plt.show()
```



Here we can see clear relationship between credit score and credit limit. Where there are levels for example, upto 650 score is getting a very minor credit limit (<1000$) where as a score between 650 to 700 is getting around 20000. Score between 700 to 750 is getting around 40K etc.

In [374…
```
# Define bin ranges
bin_ranges = [300, 450, 500, 550, 600, 650, 700, 750, 800]
```

```
# Create labels for the bins
bin_labels = [f'{start}-{end-1}' for start, end in zip(bin_ranges, bin_ranges[1:])]

# Use pd.cut to assign data to bins
df_cs_clean_1['credit_score_range'] = pd.cut(df_cs_clean_1['credit_score'], bins=bin_ranges, labels=bin_labels, include_lowest=Tru
```

In [375…

```
df_cs_clean_1.head()
```

Out[375…

	cust_id	credit_score	credit_utilisation	outstanding_debt	credit_inquiries_last_6_months	credit_limit	credit_score_range
0	1	749	0.585171	19571.0	0.0	40000.0	700-749
1	2	587	0.107928	161644.0	2.0	1250.0	550-599
2	3	544	0.854807	513.0	4.0	1000.0	500-549
3	4	504	0.336938	224.0	2.0	1000.0	500-549
4	5	708	0.586151	18090.0	2.0	40000.0	700-749

We can now see a new column called credit_score_range whcih is calculated based on the credit_score column

In [376…

```
df_cs_clean_1[['credit_score','credit_score_range', 'credit_limit']].head(3)
```

Out[376…

	credit_score	credit_score_range	credit_limit
0	749	700-749	40000.0
1	587	550-599	1250.0
2	544	500-549	1000.0

In [377…

```
df_cs_clean_1[df_cs_clean_1['credit_score_range']=="750-799"]
```

Out[377…

	cust_id	credit_score	credit_utilisation	outstanding_debt	credit_inquiries_last_6_months	credit_limit	credit_score_range
21	22	785	0.897089	36083.0	3.0	60000.0	750-799
25	26	758	0.250811	190838.0	2.0	60000.0	750-799
26	27	766	0.830908	31344.0	3.0	60000.0	750-799
29	30	798	0.222597	7238.0	2.0	60000.0	750-799
31	32	768	0.747793	35109.0	2.0	60000.0	750-799
...	...	...	...	...	...	...	...
988	985	770	0.628088	33405.0	2.0	60000.0	750-799
993	990	772	0.259958	11937.0	2.0	60000.0	750-799
996	993	782	0.477170	20305.0	2.0	60000.0	750-799
1000	997	774	0.465462	17139.0	0.0	60000.0	750-799
1003	1000	775	0.696050	33956.0	1.0	60000.0	750-799

213 rows × 7 columns

In [378…

```
df_cs_clean_1[df_cs_clean_1['credit_score_range']=="300-449"]
```

Out[378…

	cust_id	credit_score	credit_utilisation	outstanding_debt	credit_inquiries_last_6_months	credit_limit	credit_score_range
5	6	442	0.705409	246.0	4.0	500.0	300-449
11	12	429	0.627645	263.0	0.0	500.0	300-449
15	16	347	0.531660	190.0	0.0	500.0	300-449
18	19	447	0.795650	292.0	1.0	500.0	300-449
20	21	381	0.714710	307.0	0.0	500.0	300-449
...	...	...	...	...	...	...	...
981	978	371	0.435307	183.0	2.0	500.0	300-449
982	979	332	0.150815	65.0	1.0	500.0	300-449
984	981	327	0.377202	108.0	3.0	500.0	300-449
989	986	425	0.178470	56.0	4.0	500.0	300-449
998	995	360	0.594345	242.0	0.0	500.0	300-449

237 rows × 7 columns

Above you can see that for credit score range "750-799" the credit limit is 60K whereas for "300-449" it is 500. We can use MODE function to find out most frequently occuring credit limit for a given score range.

In [379…
```
mode_df = df_cs_clean_1.groupby('credit_score_range')['credit_limit'].agg(lambda x: x.mode().iloc[0]).reset_index()
mode_df
```

Out[379…

	credit_score_range	credit_limit
0	300-449	500.0
1	450-499	750.0
2	500-549	1000.0
3	550-599	1250.0
4	600-649	1500.0
5	650-699	20000.0
6	700-749	40000.0
7	750-799	60000.0

In [380…
```
df_cs_clean_1[df_cs_clean_1.credit_limit.isnull()].sample(3)
```

Out[380…

	cust_id	credit_score	credit_utilisation	outstanding_debt	credit_inquiries_last_6_months	credit_limit	credit_score_range
783	780	787	0.681978	24711.0	1.0	NaN	750-799
469	470	795	0.776496	33523.0	4.0	NaN	750-799
276	277	781	0.459545	24618.0	1.0	NaN	750-799

In [381…
```
# Merge the mode values back with the original DataFrame
df_cs_clean_2 = pd.merge(df_cs_clean_1, mode_df, on='credit_score_range', suffixes=('', '_mode'))
df_cs_clean_2.sample(3)
```

Out[381…

	cust_id	credit_score	credit_utilisation	outstanding_debt	credit_inquiries_last_6_months	credit_limit	credit_score_range	credit_limit_mode
72	526	713	0.158635	5259.0	2.0	40000.0	700-749	40000.0
37	279	741	0.352932	10846.0	1.0	NaN	700-749	40000.0
377	246	430	0.744435	263.0	2.0	500.0	300-449	500.0

In [382…
```
df_cs_clean_2[df_cs_clean_2.credit_limit.isnull()].sample(3)
```

Out[382…

	cust_id	credit_score	credit_utilisation	outstanding_debt	credit_inquiries_last_6_months	credit_limit	credit_score_range	credit_limit_mode
101	726	737	0.136048	205404.0	4.0	NaN	700-749	40000.0
693	115	619	0.128910	151.0	1.0	NaN	600-649	1500.0
485	677	300	0.785509	341.0	1.0	NaN	300-449	500.0

Above we can simple replace NaN value in credit_limit column with credit_limit_mode value. This value indicates most frequently occuring credit limit for a given credit_score_range. Hence it can be used as a replacement value.

We will create a new copy of the dataframe so that we have reproducibility and access of the older dataframe in this notebook

In [383…
```
df_cs_clean_3 = df_cs_clean_2.copy()
df_cs_clean_3['credit_limit'].fillna(df_cs_clean_3['credit_limit_mode'], inplace=True)
df_cs_clean_3.shape
```

Out[383…
```
(1000, 8)
```

In [384…
```
df_cs_clean_3.isnull().sum()
```

Out[384…
```
cust_id                         0
credit_score                    0
credit_utilisation              0
outstanding_debt                0
credit_inquiries_last_6_months  0
credit_limit                    0
credit_score_range              0
credit_limit_mode               0
dtype: int64
```

You can now see ZERO outliers in credit_limit column which means we successfully got rid of all NULL values. Hurray! 🥳

In [385…
```
df_cs_clean_3[df_cs_clean_3.cust_id==117]
```

Out[385…

	cust_id	credit_score	credit_utilisation	outstanding_debt	credit_inquiries_last_6_months	credit_limit	credit_score_range	credit_limit_mode
347	117	372	0.604427	252.0	4.0	500.0	300-449	500.0

Previously customer id 5 had null value in credit_limit. Now it has a valid value

Data Cleaning Step 3: Handle Outliers: outstanding_debt

In [386…
```
df_cs_clean_3.describe()
```

Out[386…

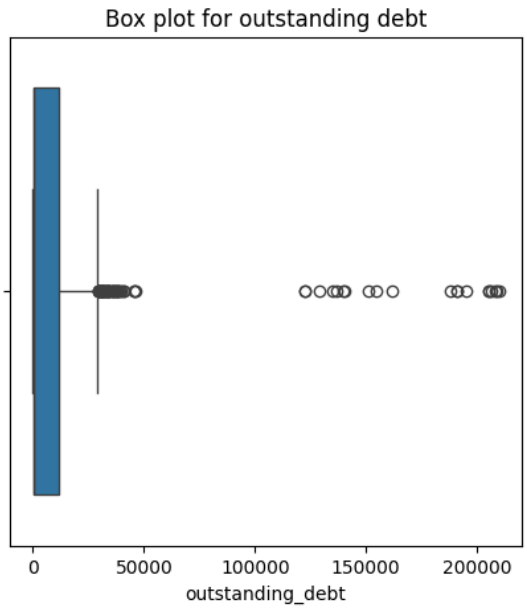
	cust_id	credit_score	credit_utilisation	outstanding_debt	credit_inquiries_last_6_months	credit_limit	credit_limit_mode
count	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000	1000.00000	1000.000000
mean	500.500000	589.182000	0.498950	9683.597000	1.955000	19733.75000	19912.500000
std	288.819436	152.284929	0.233139	25255.893671	1.414559	24717.43818	24840.914633
min	1.000000	300.000000	0.103761	33.000000	0.000000	500.00000	500.000000
25%	250.750000	460.000000	0.293917	221.000000	1.000000	750.00000	750.000000
50%	500.500000	601.500000	0.487422	550.000000	2.000000	1500.00000	1500.000000
75%	750.250000	738.000000	0.697829	11819.500000	3.000000	40000.00000	40000.000000
max	1000.000000	799.000000	0.899648	209901.000000	4.000000	60000.00000	60000.000000

When we observe min and max for various columns, we realize that outstanding_debt's max is greater than the max of credit_limit. Based on the business understanding, we know that the maximum debt that a customer can have is equal to credit limit. They would not be allowed to spend more than their credit limit. Let's see how many such cases are present in our dataset

Visualizing outliers

In [387…
```
plt.figure(figsize=(5, 5))
sns.boxplot(x=df_cs_clean_3['outstanding_debt'])
plt.title('Box plot for outstanding debt')
```

Out[387…
```
Text(0.5, 1.0, 'Box plot for outstanding debt')
```



Instead of using any statistical approach (such as standard deviation or IQR), here too we will use a business knowledge. We will mark any outstanding debt that is greater than credit limit as an outlier

In [388…
```
df_cs_clean_3[df_cs_clean_3.outstanding_debt>df_cs_clean_3.credit_limit]
```

Out[388…

	cust_id	credit_score	credit_utilisation	outstanding_debt	credit_inquiries_last_6_months	credit_limit	credit_score_range	credit_limit_mode
6	39	734	0.573023	122758.0	3.0	40000.0	700-749	40000.0
12	94	737	0.739948	137058.0	2.0	40000.0	700-749	40000.0
35	272	703	0.446886	154568.0	1.0	40000.0	700-749	40000.0
41	302	722	0.608076	122402.0	4.0	40000.0	700-749	40000.0
101	726	737	0.136048	205404.0	4.0	40000.0	700-749	40000.0
142	2	587	0.107928	161644.0	2.0	1250.0	550-599	1250.0
363	205	303	0.364360	187849.0	0.0	500.0	300-449	500.0
406	351	320	0.285081	150860.0	0.0	500.0	300-449	500.0
474	637	420	0.323984	140063.0	4.0	500.0	300-449	500.0
604	647	498	0.658087	128818.0	3.0	750.0	450-499	750.0
609	724	465	0.658173	140008.0	3.0	750.0	450-499	750.0
615	767	473	0.611750	195004.0	1.0	750.0	450-499	750.0
684	20	647	0.439132	205014.0	3.0	1500.0	600-649	1500.0
759	731	626	0.762245	209901.0	2.0	1500.0	600-649	1500.0
788	26	758	0.250811	190838.0	2.0	60000.0	750-799	60000.0
858	331	799	0.363420	208898.0	4.0	60000.0	750-799	60000.0
877	447	754	0.178394	206191.0	2.0	60000.0	750-799	60000.0
903	545	764	0.337769	135112.0	2.0	60000.0	750-799	60000.0
930	699	775	0.385100	190717.0	2.0	60000.0	750-799	60000.0
968	863	792	0.399555	208406.0	3.0	60000.0	750-799	60000.0

We will replace these outliers with credit_limit. We can assume that there was some data processing error due to we got these high numbers and it is ok to replace them with a credit_limit

In [389…

```
df_cs_clean_3.loc[df_cs_clean_3['outstanding_debt'] > df_cs_clean_3['credit_limit'], 'outstanding_debt']
```

Out[389…

```
6      122758.0
12     137058.0
35     154568.0
41     122402.0
101    205404.0
142    161644.0
363    187849.0
406    150860.0
474    140063.0
604    128818.0
609    140008.0
615    195004.0
684    205014.0
759    209901.0
788    190838.0
858    208898.0
877    206191.0
903    135112.0
930    190717.0
968    208406.0
Name: outstanding_debt, dtype: float64
```

In [390…

```
df_cs_clean_3.loc[df_cs_clean_3['outstanding_debt'] > df_cs_clean_3['credit_limit'], 'outstanding_debt'] = df_cs_clean_3['credit_l
```

In [391…

```
df_cs_clean_3.loc[[55,66]]
```

Out[391…

	cust_id	credit_score	credit_utilisation	outstanding_debt	credit_inquiries_last_6_months	credit_limit	credit_score_range	credit_limit_mode
55	408	737	0.354976	7223.0	4.0	40000.0	700-749	40000.0
66	489	743	0.195717	6660.0	4.0	40000.0	700-749	40000.0

In [392…

```
df_cs_clean_3[df_cs_clean_3.outstanding_debt>df_cs_clean_3.credit_limit]
```

Out[392…

cust_id	credit_score	credit_utilisation	outstanding_debt	credit_inquiries_last_6_months	credit_limit	credit_score_range	credit_limit_mode

All outliers in column outstanding_debt are now GONE. Hurray 🙌🏻🤪

In [393…

```
df_cs_clean_3.describe()
```

Out[393…

	cust_id	credit_score	credit_utilisation	outstanding_debt	credit_inquiries_last_6_months	credit_limit	credit_limit_mode
count	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000	1000.00000	1000.000000
mean	500.500000	589.182000	0.498950	6850.084000	1.955000	19733.75000	19912.500000
std	288.819436	152.284929	0.233139	10683.473561	1.414559	24717.43818	24840.914633
min	1.000000	300.000000	0.103761	33.000000	0.000000	500.00000	500.000000
25%	250.750000	460.000000	0.293917	221.000000	1.000000	750.00000	750.000000
50%	500.500000	601.500000	0.487422	541.500000	2.000000	1500.00000	1500.000000
75%	750.250000	738.000000	0.697829	10924.500000	3.000000	40000.00000	40000.000000
max	1000.000000	799.000000	0.899648	60000.000000	4.000000	60000.00000	60000.000000

Data Exploration: Visualizing Correlation in Credit Score Table

In [394…

```
df_cust.head(2)
```

Out[394…

	cust_id	name	gender	age	location	occupation	annual_income	marital_status	age_group
0	1	Dhruv Bansal	Female	51.0	City	Business Owner	358211.0	Married	49-65
1	2	Kabir Thakur	Female	47.0	City	Consultant	65172.0	Single	26-48

In [395…

```
df_cs_clean_3.head(2)
```

Out[395…

	cust_id	credit_score	credit_utilisation	outstanding_debt	credit_inquiries_last_6_months	credit_limit	credit_score_range	credit_limit_mode
0	1	749	0.585171	19571.0	0.0	40000.0	700-749	40000.0
1	5	708	0.586151	18090.0	2.0	40000.0	700-749	40000.0

In [396…

```
df_merged = df_cust.merge(df_cs_clean_3, on='cust_id', how='inner')
df_merged.head(2)
```

Out[396…

	cust_id	name	gender	age	location	occupation	annual_income	marital_status	age_group	credit_score	credit_utilisation	outstanding_c
0	1	Dhruv Bansal	Female	51.0	City	Business Owner	358211.0	Married	49-65	749	0.585171	195
1	2	Kabir Thakur	Female	47.0	City	Consultant	65172.0	Single	26-48	587	0.107928	12

In [397…

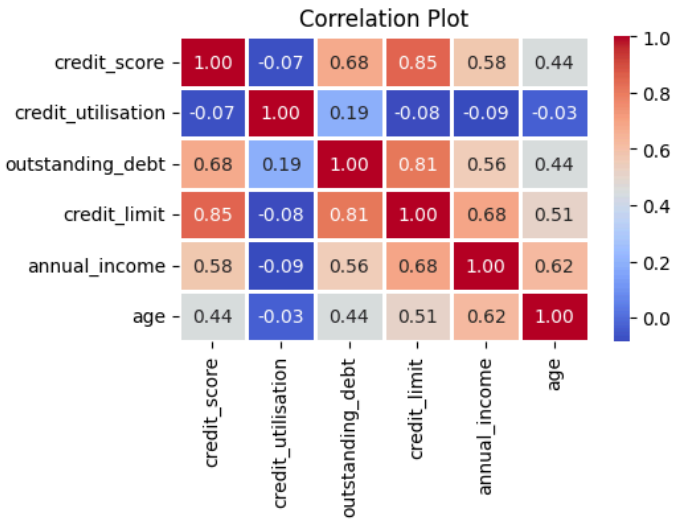
```
numerical_cols = ['credit_score', 'credit_utilisation', 'outstanding_debt', 'credit_limit', 'annual_income','age']

correlation_matrix = df_merged[numerical_cols].corr()
correlation_matrix
```

Out[397…

	credit_score	credit_utilisation	outstanding_debt	credit_limit	annual_income	age
credit_score	1.000000	-0.070445	0.680654	0.847952	0.575751	0.444917
credit_utilisation	-0.070445	1.000000	0.192838	-0.080493	-0.086368	-0.027713
outstanding_debt	0.680654	0.192838	1.000000	0.810581	0.555661	0.444301
credit_limit	0.847952	-0.080493	0.810581	1.000000	0.684775	0.510993
annual_income	0.575751	-0.086368	0.555661	0.684775	1.000000	0.619037
age	0.444917	-0.027713	0.444301	0.510993	0.619037	1.000000

In [398…

```
# Create a heatmap of the correlation matrix
plt.figure(figsize=(5, 3))
sns.heatmap(correlation_matrix, annot=True, fmt=".2f", cmap='coolwarm', linewidths=0.8)
plt.title('Correlation Plot')
plt.show()
```



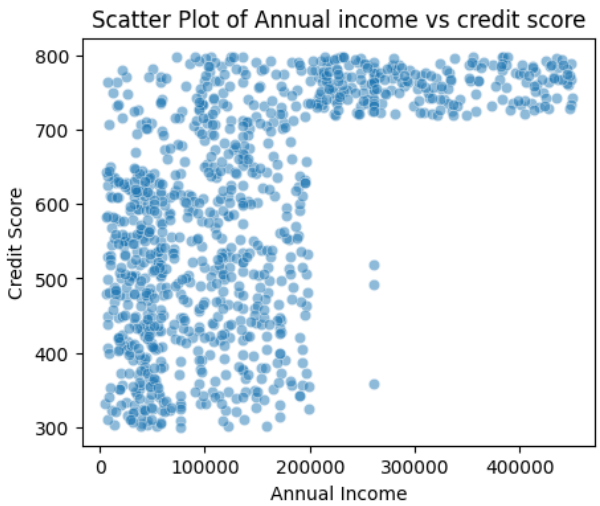
You can see a high correlation between credit limit and credit score (~0.85)

Also credit limit and annual income has a high correlation.

This correlation table can be used for further analysis. It shows if one variable has relationship with the other variable

In [399…

```
# Just looking if there is any relation between annual_income and credit score
plt.figure(figsize=(5, 4))
sns.scatterplot(x='annual_income', y='credit_score', data=df_merged, alpha=0.5)
plt.title('Scatter Plot of Annual income vs credit score')
plt.xlabel('Annual Income')
plt.ylabel('Credit Score')
plt.show()
```



No clear pattern observed

Transactions Table

In [400…

```
df_trans.head(2)
```

Out[400…

	tran_id	cust_id	tran_date	tran_amount	platform	product_category	payment_type
0	1	705	2023-01-01	63	Flipkart	Electronics	Phonepe
1	2	385	2023-01-01	99	Alibaba	Fashion & Apparel	Credit Card

In [401…

```
df_trans.shape
```

Out[401…

```
(500000, 7)
```

Data Cleaning Step 1: Handle NULL Values: platform column

In [403…

```
df_trans.isnull().sum()
```

Out[403…

```
tran_id                0
cust_id                0
tran_date              0
tran_amount            0
platform            4941
product_category       0
payment_type           0
dtype: int64
```

platform has a lot of null values. Let's check them further

In [404…

```
df_trans[df_trans.platform.isnull()]
```

Out[404…

	tran_id	cust_id	tran_date	tran_amount	platform	product_category	payment_type
355	356	58	2023-01-01	237	None	Electronics	Net Banking
418	419	383	2023-01-01	338	None	Electronics	Credit Card
607	608	421	2023-01-01	700	None	Electronics	Phonepe
844	845	945	2023-01-01	493	None	Sports	Credit Card
912	913	384	2023-01-01	85	None	Fashion & Apparel	Phonepe
...	...	...	...	...	...	...	...
499579	499580	924	2023-09-05	31	None	Fashion & Apparel	Gpay
499646	499647	944	2023-09-05	58445	None	Fashion & Apparel	Phonepe
499725	499726	620	2023-09-05	15	None	Sports	Net Banking
499833	499834	616	2023-09-05	97	None	Fashion & Apparel	Credit Card
499997	499998	57	2023-09-05	224	None	Garden & Outdoor	Phonepe

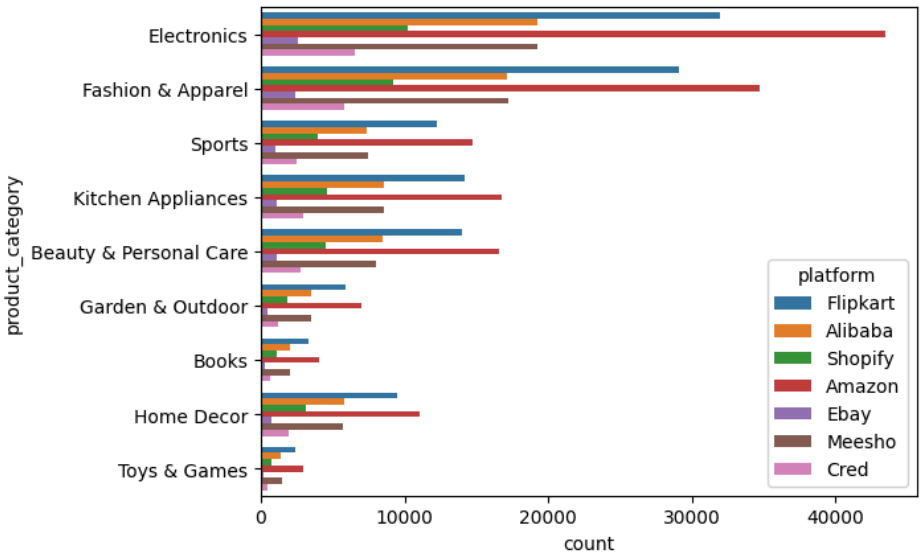
4941 rows × 7 columns

In [405…

```
sns.countplot(y='product_category', hue='platform', data=df_trans)
```

Out[405…

```
<Axes: xlabel='count', ylabel='product_category'>
```



In the above chart, you can see that in all product categories Amazon is the platform that is used the most for making purchases. For handling null values in platform may be we can just replace them using "Amazon" as a product platform just because it is used most frequently

In [406…

```
df_trans.platform.mode()
```

Out[406…

```
0    Amazon
Name: platform, dtype: object
```

In [407…

```
df_trans.platform.mode()[0]
```

Out[407…

```
'Amazon'
```

In [408…

```
df_trans['platform'].fillna(df_trans.platform.mode()[0], inplace=True)
```

In [409…

```
df_trans.isnull().sum()
```

Out[409…
```
tran_id             0
cust_id             0
tran_date           0
tran_amount         0
platform            0
product_category    0
payment_type        0
dtype: int64
```

Once again we got rid of NULL values

Data Cleaning Step 2: Treat Outliers: tran_amount

In [410…
```
df_trans.describe()
```

Out[410…

	tran_id	cust_id	tran_amount
count	500000.000000	500000.000000	500000.00000
mean	250000.500000	501.400428	3225.20733
std	144337.711635	288.641924	13098.74276
min	1.000000	1.000000	0.00000
25%	125000.750000	252.000000	64.00000
50%	250000.500000	502.000000	141.00000
75%	375000.250000	752.000000	397.00000
max	500000.000000	1000.000000	69999.00000

We can see transactions with 0 amount. These seem to be invalid

In [280…
```
df_trans_zero = df_trans[df_trans.tran_amount==0]
df_trans_zero.head(3)
```

Out[280…

	tran_id	cust_id	tran_date	tran_amount	platform	product_category	payment_type
120	121	440	2023-01-01	0	Amazon	Electronics	Credit Card
141	142	839	2023-01-01	0	Amazon	Electronics	Credit Card
517	518	147	2023-01-01	0	Amazon	Electronics	Credit Card

In [281…
```
df_trans_zero.shape
```

Out[281…
```
(4734, 7)
```

In [282…
```
df_trans_zero.platform.value_counts()
```

Out[282…
```
platform
Amazon    4734
Name: count, dtype: int64
```

In [283…
```
df_trans_zero[['platform','product_category','payment_type']].value_counts()
```

Out[283…
```
platform  product_category  payment_type
Amazon    Electronics       Credit Card     4734
Name: count, dtype: int64
```

It appears that when platform=Amazon, product_category=Eletronics and payment_type=Credit Card, at that time we get all these zero transactions. We need to find other transactions in this group and find its median to replace these zero values. We are not using mean because we can see some outliers as well in this column

In [284…
```
df_trans_1 = df_trans[(df_trans.platform=='Amazon')&(df_trans.product_category=="Electronics")&(df_trans.payment_type=="Credit Car
df_trans_1.shape
```

Out[284…
```
(15288, 7)
```

In [285…
```
df_trans_1[df_trans_1.tran_amount>0]
```

Out[285…

	tran_id	cust_id	tran_date	tran_amount	platform	product_category	payment_type
109	110	887	2023-01-01	635	Amazon	Electronics	Credit Card
173	174	676	2023-01-01	60439	Amazon	Electronics	Credit Card
190	191	763	2023-01-01	697	Amazon	Electronics	Credit Card
263	264	528	2023-01-01	421	Amazon	Electronics	Credit Card
311	312	936	2023-01-01	537	Amazon	Electronics	Credit Card
...	...	...	...	...	...	...	...
499766	499767	723	2023-09-05	909	Amazon	Electronics	Credit Card
499793	499794	586	2023-09-05	304	Amazon	Electronics	Credit Card
499812	499813	688	2023-09-05	425	Amazon	Electronics	Credit Card
499860	499861	373	2023-09-05	480	Amazon	Electronics	Credit Card
499885	499886	520	2023-09-05	643	Amazon	Electronics	Credit Card

10554 rows × 7 columns

In [286…

```
median_to_replace = df_trans_1[df_trans_1.tran_amount>0].tran_amount.median()
median_to_replace
```

Out[286…

```
554.0
```

In [287…

```
df_trans['tran_amount'].replace(0,median_to_replace, inplace=True)
```

In [288…

```
df_trans[df_trans.tran_amount==0]
```

Out[288…

tran_id	cust_id	tran_date	tran_amount	platform	product_category	payment_type

As you can see above, no zero values are left in tran_amount column

In [289…

```
df_trans.tran_amount.describe()
```

Out[289…

```
count    500000.000000
mean       3230.452602
std       13097.561071
min           2.000000
25%          66.000000
50%         146.000000
75%         413.000000
max       69999.000000
Name: tran_amount, dtype: float64
```

In [290…

```
df_trans[df_trans['tran_amount']<1000].describe()
```

Out[290…

	tran_id	cust_id	tran_amount
count	475000.000000	475000.000000	475000.000000
mean	250041.699922	501.375499	240.667608
std	144285.259913	288.606185	244.487110
min	1.000000	1.000000	2.000000
25%	125126.750000	252.000000	63.000000
50%	250100.500000	502.000000	131.000000
75%	374928.250000	751.000000	348.000000
max	500000.000000	1000.000000	999.000000

In [165…

```
Q1, Q3 = df_trans['tran_amount'].quantile([0.25, 0.75])
IQR = Q3 - Q1
lower = Q1 - 2 * IQR
upper = Q3 + 2 * IQR

lower, upper
```

Out[165…

```
(-628.0, 1107.0)
```

In [166…

```
df_trans[df_trans.tran_amount<upper].tran_amount.max()
```

Out[166…

```
999
```

In [167…

```
df_trans[df_trans.tran_amount>upper].tran_amount.min()
```

Out[167…

```
50000
```

In [168…

```
df_trans_outliers = df_trans[df_trans.tran_amount>=upper]
```

```
df_trans_outliers
```

Out[168…

	tran_id	cust_id	tran_date	tran_amount	platform	product_category	payment_type
26	27	380	2023-01-01	61963	Shopify	Beauty & Personal Care	Credit Card
49	50	287	2023-01-01	57869	Amazon	Toys & Games	Gpay
94	95	770	2023-01-01	52881	Ebay	Kitchen Appliances	Credit Card
104	105	549	2023-01-01	58574	Flipkart	Fashion & Apparel	Gpay
113	114	790	2023-01-01	51669	Shopify	Kitchen Appliances	Credit Card
...	...	...	...	...	...	...	...
499742	499743	868	2023-09-05	55131	Meesho	Fashion & Apparel	Gpay
499888	499889	614	2023-09-05	59679	Meesho	Fashion & Apparel	Net Banking
499900	499901	811	2023-09-05	60184	Flipkart	Sports	Debit Card
499966	499967	662	2023-09-05	54678	Meesho	Sports	Gpay
499996	499997	569	2023-09-05	53022	Meesho	Fashion & Apparel	Net Banking

25000 rows × 7 columns

In [169…

```
df_trans_normal = df_trans[df_trans.tran_amount<upper]
df_trans_normal
```

Out[169…

	tran_id	cust_id	tran_date	tran_amount	platform	product_category	payment_type
0	1	705	2023-01-01	63	Flipkart	Electronics	Phonepe
1	2	385	2023-01-01	99	Alibaba	Fashion & Apparel	Credit Card
2	3	924	2023-01-01	471	Shopify	Sports	Phonepe
3	4	797	2023-01-01	33	Shopify	Fashion & Apparel	Gpay
4	5	482	2023-01-01	68	Amazon	Fashion & Apparel	Net Banking
...	...	...	...	...	...	...	...
499994	499995	679	2023-09-05	59	Ebay	Beauty & Personal Care	Gpay
499995	499996	791	2023-09-05	43	Amazon	Books	Phonepe
499997	499998	57	2023-09-05	224		Garden & Outdoor	Phonepe
499998	499999	629	2023-09-05	538	Flipkart	Home Decor	Gpay
499999	500000	392	2023-09-05	346	Amazon	Kitchen Appliances	Net Banking

475000 rows × 7 columns

In [170…

```
tran_mean_per_category = df_trans_normal.groupby("product_category")["tran_amount"].mean()
tran_mean_per_category
```

Out[170…

```
product_category
Beauty & Personal Care     92.167205
Books                      29.553515
Electronics               510.172685
Fashion & Apparel          64.553463
Garden & Outdoor          125.630277
Home Decor                302.487561
Kitchen Appliances        176.773288
Sports                    269.181631
Toys & Games               50.333298
Name: tran_amount, dtype: float64
```

In [171…

```
df_trans.loc[df_trans_outliers.index]
```

Out[171…

	tran_id	cust_id	tran_date	tran_amount	platform	product_category	payment_type
26	27	380	2023-01-01	61963	Shopify	Beauty & Personal Care	Credit Card
49	50	287	2023-01-01	57869	Amazon	Toys & Games	Gpay
94	95	770	2023-01-01	52881	Ebay	Kitchen Appliances	Credit Card
104	105	549	2023-01-01	58574	Flipkart	Fashion & Apparel	Gpay
113	114	790	2023-01-01	51669	Shopify	Kitchen Appliances	Credit Card
...	...	...	...	...	...	...	...
499742	499743	868	2023-09-05	55131	Meesho	Fashion & Apparel	Gpay
499888	499889	614	2023-09-05	59679	Meesho	Fashion & Apparel	Net Banking
499900	499901	811	2023-09-05	60184	Flipkart	Sports	Debit Card
499966	499967	662	2023-09-05	54678	Meesho	Sports	Gpay
499996	499997	569	2023-09-05	53022	Meesho	Fashion & Apparel	Net Banking

25000 rows × 7 columns

In [172…

```
df_trans.loc[df_trans_outliers.index, 'tran_amount'] = df_trans_outliers['product_category'].map(tran_mean_per_category)
```

In [173…

```
df_trans.loc[df_trans_outliers.index]
```

Out[173…

	tran_id	cust_id	tran_date	tran_amount	platform	product_category	payment_type
26	27	380	2023-01-01	92.167205	Shopify	Beauty & Personal Care	Credit Card
49	50	287	2023-01-01	50.333298	Amazon	Toys & Games	Gpay
94	95	770	2023-01-01	176.773288	Ebay	Kitchen Appliances	Credit Card
104	105	549	2023-01-01	64.553463	Flipkart	Fashion & Apparel	Gpay
113	114	790	2023-01-01	176.773288	Shopify	Kitchen Appliances	Credit Card
...	...	...	...	...	...	...	...
499742	499743	868	2023-09-05	64.553463	Meesho	Fashion & Apparel	Gpay
499888	499889	614	2023-09-05	64.553463	Meesho	Fashion & Apparel	Net Banking
499900	499901	811	2023-09-05	269.181631	Flipkart	Sports	Debit Card
499966	499967	662	2023-09-05	269.181631	Meesho	Sports	Gpay
499996	499997	569	2023-09-05	64.553463	Meesho	Fashion & Apparel	Net Banking

25000 rows × 7 columns

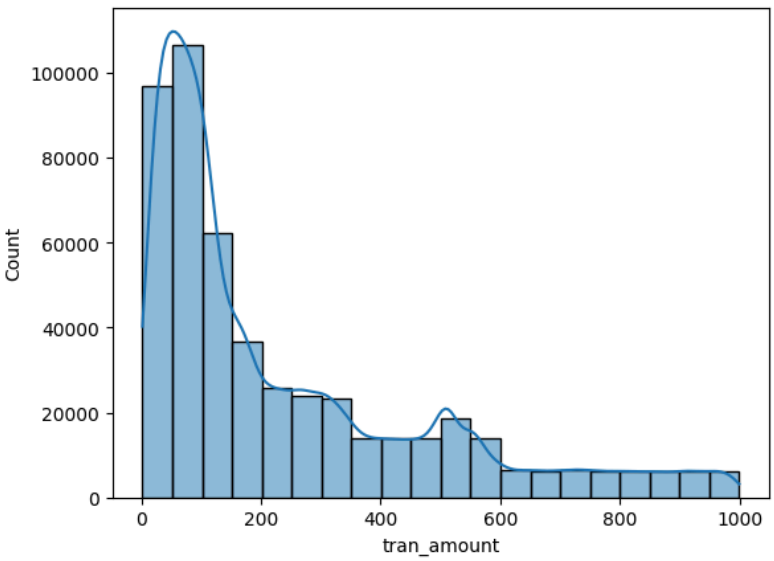
You can now see that we got rid of outliers from tran_amount column. Great job folks 👍🏻 🔥 🔥

In [174…

```
sns.histplot(x='tran_amount', data=df_trans, bins=20, kde=True)
```

Out[174…

```
<Axes: xlabel='tran_amount', ylabel='Count'>
```



Above shows the histogram of transactions after the removal of outliers. You can see that distribution is right skewed. Transaction amount now is less than 1000

Data Visualization: Payment Type Distribution

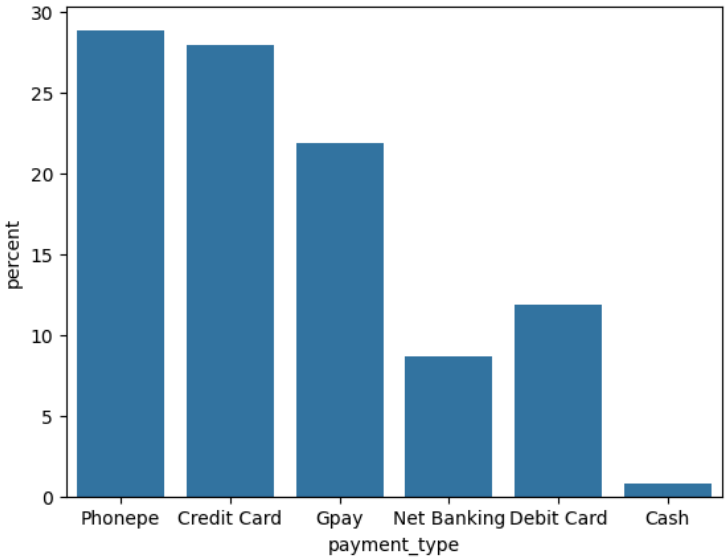
In [422…
```
df_trans.head(3)
```

Out[422…

	tran_id	cust_id	tran_date	tran_amount	platform	product_category	payment_type
0	1	705	2023-01-01	63	Flipkart	Electronics	Phonepe
1	2	385	2023-01-01	99	Alibaba	Fashion & Apparel	Credit Card
2	3	924	2023-01-01	471	Shopify	Sports	Phonepe

In [415…
```
sns.countplot(x=df_trans.payment_type, stat='percent')
```

Out[415…
```
<Axes: xlabel='payment_type', ylabel='percent'>
```



Distribution of payment types across age groups

In [416…
```
df_merged_2 = df_merged.merge(df_trans, on='cust_id', how='inner')
df_merged_2.head(3)
```

Out[416…

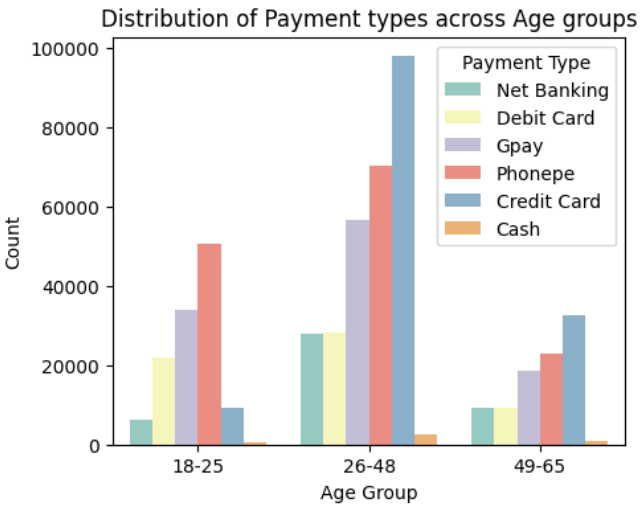
	cust_id	name	gender	age	location	occupation	annual_income	marital_status	age_group	credit_score	…	credit_inquiries_last_6_month
0	1	Dhruv Bansal	Female	51.0	City	Business Owner	358211.0	Married	49-65	749	…	0.
1	1	Dhruv Bansal	Female	51.0	City	Business Owner	358211.0	Married	49-65	749	…	0.
2	1	Dhruv Bansal	Female	51.0	City	Business Owner	358211.0	Married	49-65	749	…	0.

3 rows × 22 columns

In [417…
```
df_merged_2.shape
```

Out[417…
```
(500000, 22)
```

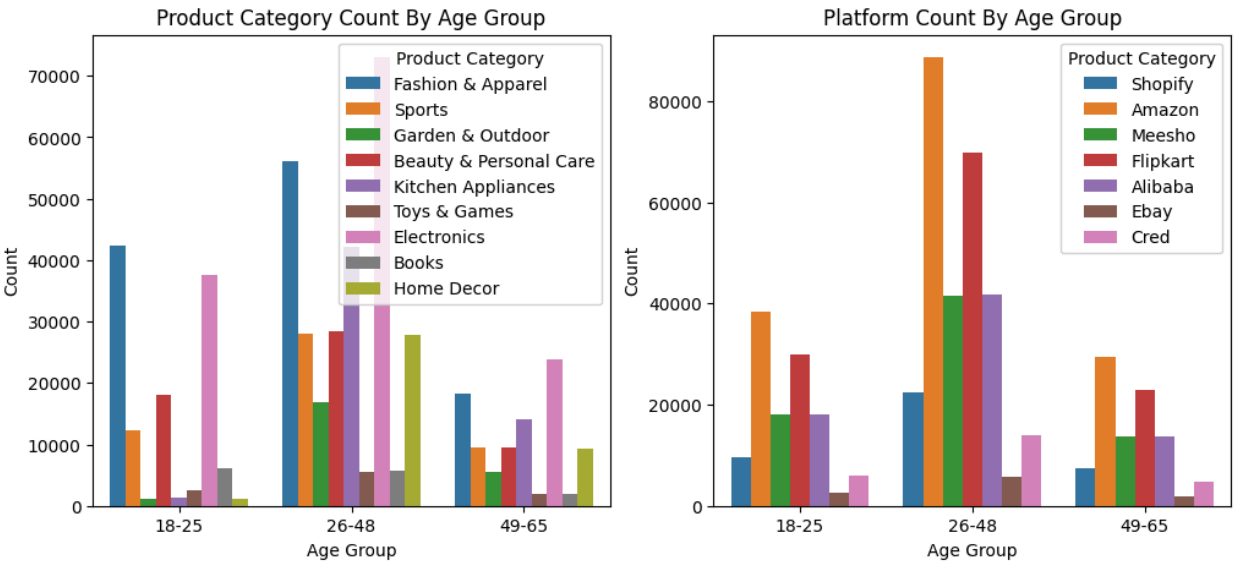
In [418…
```
plt.figure(figsize=(5, 4))
sns.countplot(x='age_group', hue='payment_type', data=df_merged_2, palette='Set3')
plt.title('Distribution of Payment types across Age groups')
plt.xlabel('Age Group')
plt.ylabel('Count')
plt.xticks(rotation=0)
plt.legend(title='Payment Type', loc='upper right')

plt.show()
```



From above analysis, we can see that age group 18-25 has less exposure to credit cards compared to other groups

In [419…

```
fig, (ax1, ax2) = plt.subplots(1,2, figsize=(12,5))

sns.countplot(x='age_group', hue="product_category", data=df_merged_2, ax=ax1)
ax1.set_title("Product Category Count By Age Group")
ax1.set_xlabel("Age Group")
ax1.set_ylabel("Count")
ax1.legend(title="Product Category", loc='upper right')

sns.countplot(x='age_group', hue="platform", data=df_merged_2, ax=ax2)
ax2.set_title("Platform Count By Age Group")
ax2.set_xlabel("Age Group")
ax2.set_ylabel("Count")
ax2.legend(title="Product Category", loc='upper right')

plt.show()
```



Observations:

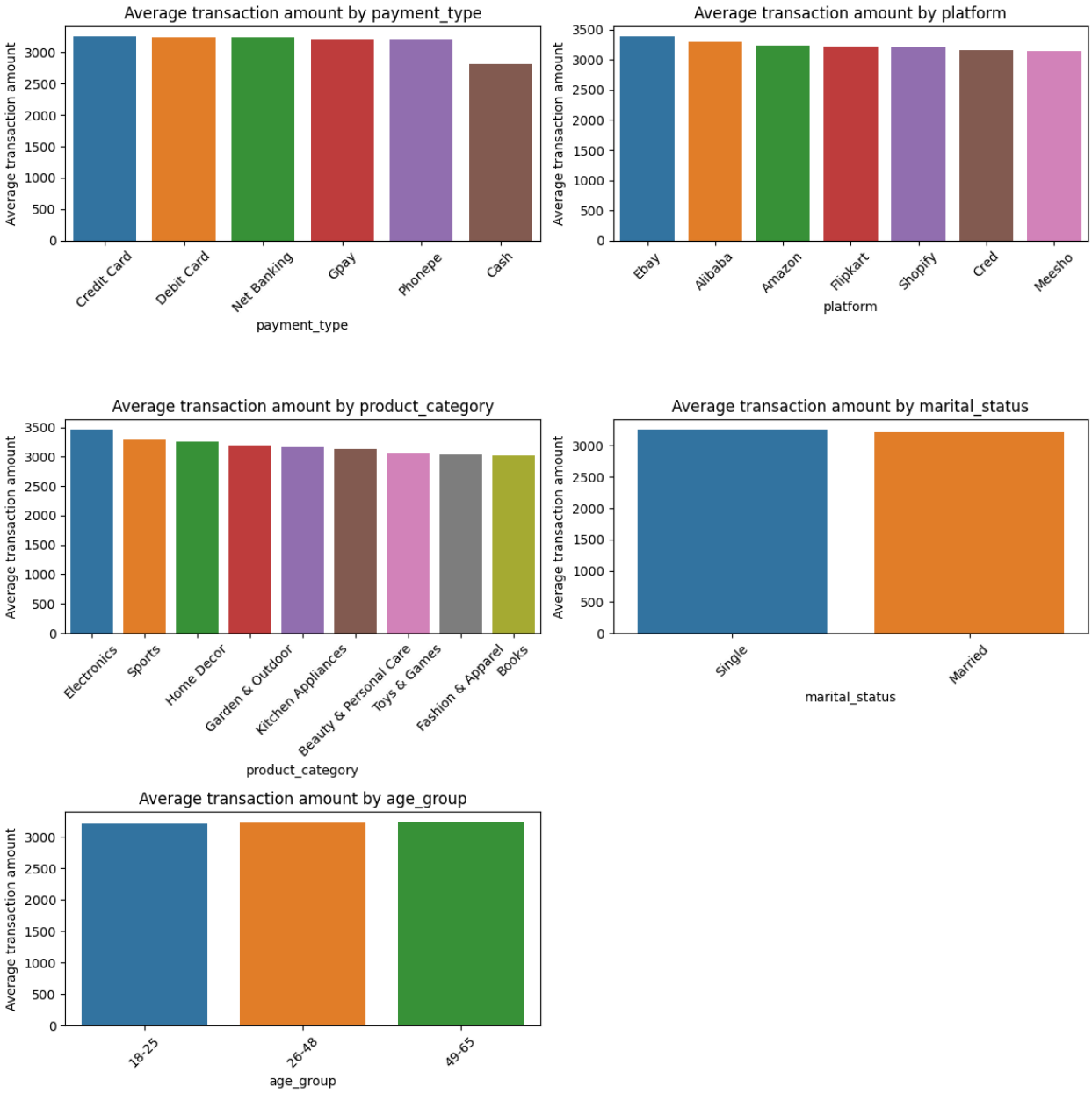
1. Top 3 purchasing categories of customers in age group (18 -25) : Electronics, Fashion & Apparel, Beauty & personal care
2. Top platforms : Amazon, Flipkart, Alibaba

Data Visualization: Average Transaction Amount

In [421…

```
# List of categorical columns
cat_cols = ['payment_type', 'platform', 'product_category', 'marital_status', 'age_group']

num_rows = 3
# Create subplots
fig, axes = plt.subplots(num_rows, 2, figsize=(12, 4 * num_rows))

# Flatten the axes array to make it easier to iterate
axes = axes.flatten()

# Create subplots for each categorical column
for i, cat_col in enumerate(cat_cols):
    # Calculate the average annual income for each category
```

```
    avg_tran_amount_by_category = df_merged_2.groupby(cat_col)['tran_amount'].mean().reset_index()

    # Sort the data by 'annual_income' before plotting
    sorted_data = avg_tran_amount_by_category.sort_values(by='tran_amount', ascending=False)

    sns.barplot(x=cat_col, y='tran_amount', data=sorted_data, ci=None, ax=axes[i], palette='tab10')
    axes[i].set_title(f'Average transaction amount by {cat_col}')
    axes[i].set_xlabel(cat_col)
    axes[i].set_ylabel('Average transaction amount')

    # Rotate x-axis labels for better readability
    axes[i].set_xticklabels(axes[i].get_xticklabels(), rotation=45)

# Hide any unused subplots
for i in range(len(cat_cols), len(axes)):
    fig.delaxes(axes[i])
plt.tight_layout()
plt.show()
```



In [131…

```
df_trans.describe()
```

Out[131…

	tran_id	cust_id	tran_amount
count	500000.000000	500000.000000	500000.000000
mean	250000.500000	500.577940	241.473582
std	144337.711635	288.739056	243.267746
min	1.000000	1.000000	2.000000
25%	125000.750000	250.000000	64.607934
50%	250000.500000	501.000000	132.000000
75%	375000.250000	750.000000	351.000000
max	500000.000000	1000.000000	999.000000

Further Analysis On Age Group

Let us do further analysis on age group to figure out their average income, credit limit, credit score etc

In [429…

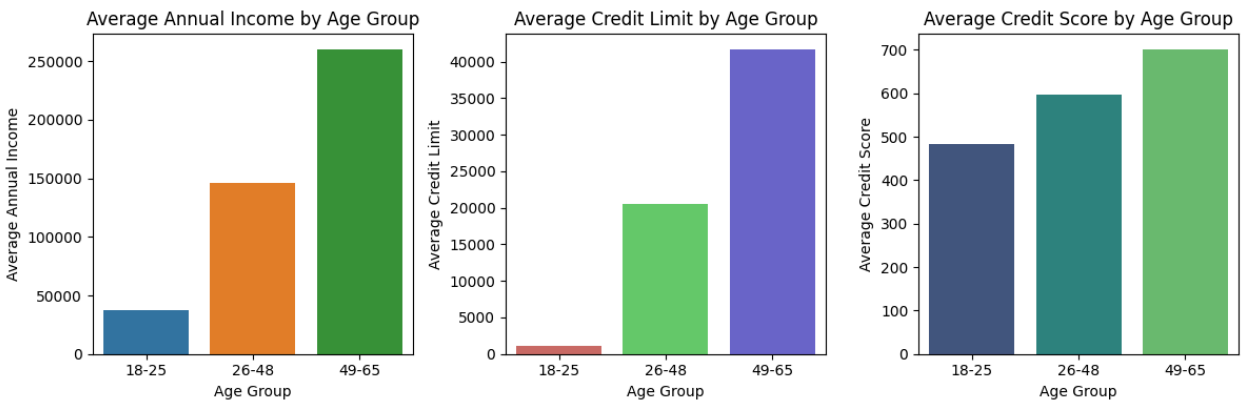
```
# Group the data by age group and calculate the average credit_limit and credit_score
age_group_metrics = df_merged.groupby('age_group')[['annual_income', 'credit_limit', 'credit_score']].mean().reset_index()
age_group_metrics
```

Out[429…

	age_group	annual_income	credit_limit	credit_score
0	18-25	37091.235772	1130.081301	484.451220
1	26-48	145869.623457	20560.846561	597.569665
2	49-65	260165.925134	41699.197861	701.524064

In [439…

```
# Create subplots
fig, (ax1, ax2, ax3) = plt.subplots(1, 3, figsize=(12, 4))

# Plot 1: Average annual income by age group
sns.barplot(x='age_group', y='annual_income', data=age_group_metrics, palette='tab10', ax=ax1)
ax1.set_title('Average Annual Income by Age Group')
ax1.set_xlabel('Age Group')
ax1.set_ylabel('Average Annual Income')
ax1.tick_params(axis='x', rotation=0)

# Plot 2: Average Max Credit Limit by Age Group
sns.barplot(x='age_group', y='credit_limit', data=age_group_metrics, palette='hls', ax=ax2)
ax2.set_title('Average Credit Limit by Age Group')
ax2.set_xlabel('Age Group')
ax2.set_ylabel('Average Credit Limit')
ax2.tick_params(axis='x', rotation=0)

# Plot 3: Average Credit Score by Age Group
sns.barplot(x='age_group', y='credit_score', data=age_group_metrics, palette='viridis', ax=ax3)
ax3.set_title('Average Credit Score by Age Group')
ax3.set_xlabel('Age Group')
ax3.set_ylabel('Average Credit Score')
ax3.tick_params(axis='x', rotation=0)

plt.tight_layout()
plt.show()
```



Finalize Target Market For a Trial Credit Card Launch

Targeting Untapped market

1. People with age group of 18 -25 accounts to ~26% of customer base in the data

2. Avg annual income of this group is less than 50k
3. They don't have much credit history which is getting reflected in their credit score and credit limit
4. Usage of credit cards as payment type is relatively low compared to other groups
5. Top 3 most shopping products categories : Electronics, Fashion & Apparel, Beauty & Personal care